Geometric Representation

ECE/MAE 5930 Mobile Robotics

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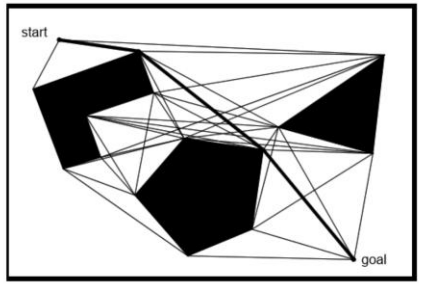
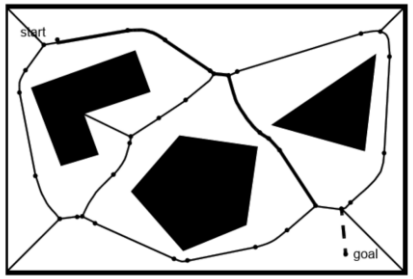
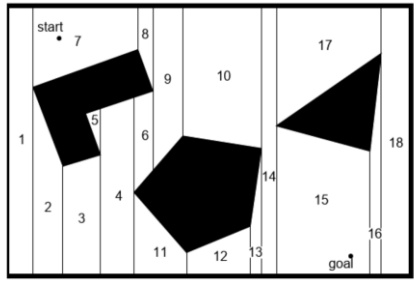
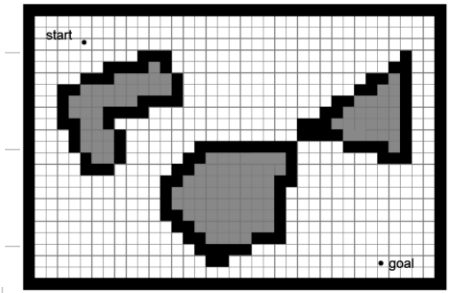
Motivation

- Previously, we looked at discrete planning: How to get from point A to point B
  - What are we planning over: graph









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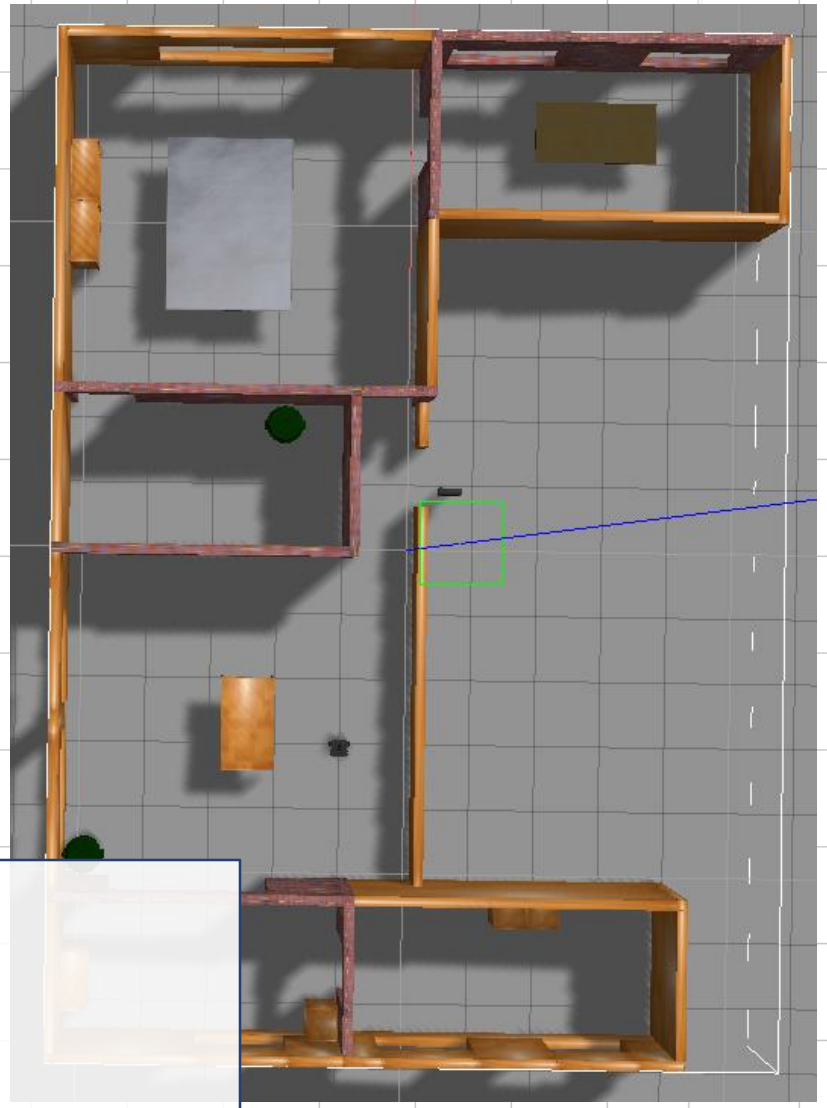
  - How do we plan:
    - Pure feasible (breadth, depth), optimal (Dijkstra, A*), number of variations
  - Underlying assumptions in all of our previous discussion:
    - Vehicle being a point
    - Working in a global frame
- This section we discuss Geometric Representation
  - How to represent all of the points in a vehicle
  - Coordinate transforms (where is (0,0) and how are the axes oriented)
- Today: polygon-based representation of world and robot

How do we model the world?

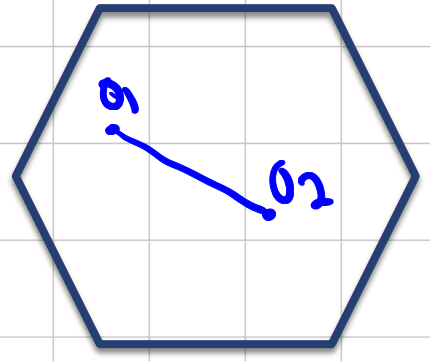


Define World:

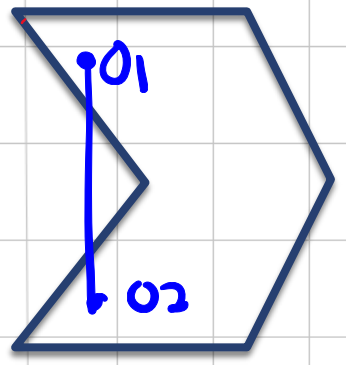
- $W \subseteq \mathbb{R}^2$ or $W \subseteq \mathbb{R}^3$
- Two types of entities
  - Obstacles (stationary) $\rightarrow W_{obs}, O \subseteq W$
  - Robot/moving obstacles $\rightarrow R(t) \subseteq W$

Polygonal Models

- Objects defined in terms of finite set of **unions**, **intersections**, and **set differences**
- Utilize convex polygon definition



Convex



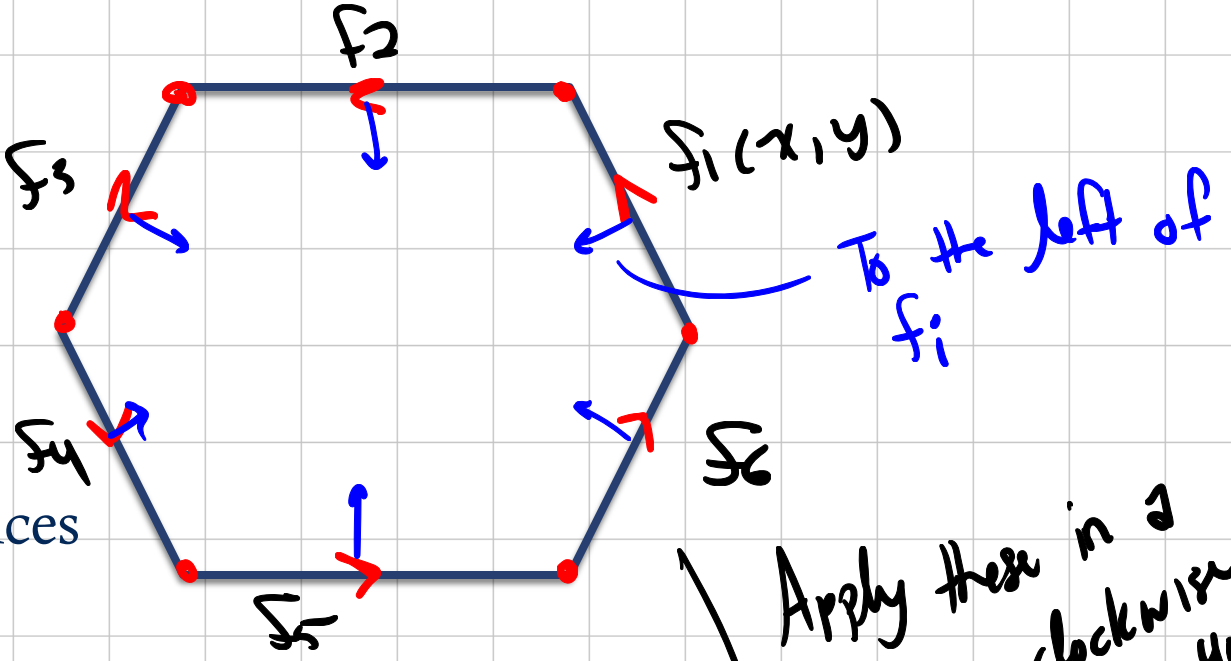
Non-convex

$o_1 \in O \quad o_2 \in O, \lambda \in [0\ 1]$

convex $\Rightarrow (o_1 x + (1-\lambda) o_2) \in O$

Polygon Boundary Representation

- Definition
  - Vertex: corner
  - Edge: line segment
- Boundary representation
  - Defined using vertices and edges
  - Polygon specified by vertex sequences



Apply these in a counter clockwise form gives you a system of EQ's

- Solid representation: Intersection of m half-planes
  - Half-plane: set of all points lying to one side of a line

$$(x_2 - x_1)y = (y_2 - y_1)x + b(x_2 - x_1) \Rightarrow 0 = \underbrace{(x_2 - x_1)y + (y_2 - y_1)x + b(x_2 - x_1)}_{f(x,y)}$$

If we are on the line $f(x,y) = 0$ where

Generally H_i is associated with f_i

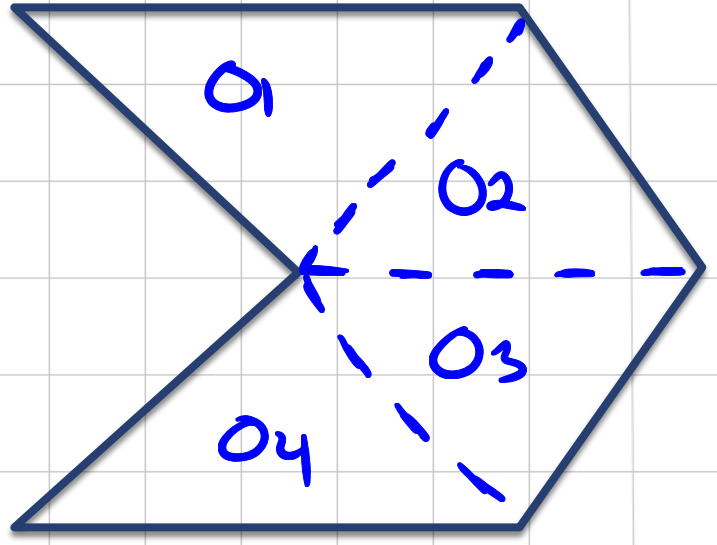
$$H_i \{(x,y) \mid f_i(x,y) \leq 0\}$$

- Polygon is the intersection of all the half plane

$$0 = \underbrace{(x_1 - x_2)}_{a} y + \underbrace{(y_2 - y_1)}_{b} x + (\underbrace{-ax_1 - by_1})$$

Non-Convex Polygons

- Idea: decompose into convex polygons
  - Take the union
  - Note: decomposition is not unique



$$O = O_1 \cup O_2 \cup O_3 \cup O_4$$

Aside – Defining Predicates

- And / or operators

$a \wedge b \rightarrow$ and

$a \vee b \rightarrow$ or

- Logical predicate:

$\varnothing : W \rightarrow \{\text{true}, \text{false}\}$

Halfspace (H_i) $e_i(x,y) \begin{cases} \text{true} & f_i(x,y) \leq 0 \\ \text{false} & f_i(x,y) > 0 \end{cases}$

inside polygon

outside polygon

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Using Predicates for a Single Convex Obstacle

- Let

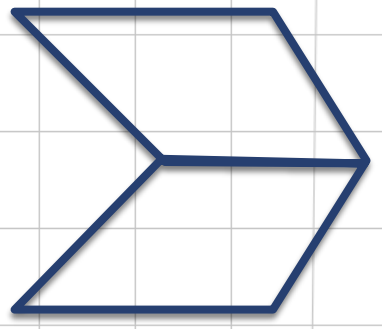
- Define whether point lies within an obstacle using combination of predicates

Obstacle 1

$$\alpha_i(x,y) = e_1(x,y) \wedge e_2(x,y) \cdots \wedge e_m(x,y)$$

- Look at entire obstacle region

$$\emptyset(x,y) = \alpha_1 \vee \alpha_2 \vee \alpha_3 \cdots \vee \alpha_n$$



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Going to 3D

- Polygons → Polyhedra
- Boundary definition
  - Vertices & edges → Vertices, edges, & faces
- Solid definition
  - Half-plane → Half-space

$$f_i(x,y) = ax + by + cz + d$$

$$H_i = \{(x,y,z) \in W \mid f_i(x,y,z) \leq 0\}$$

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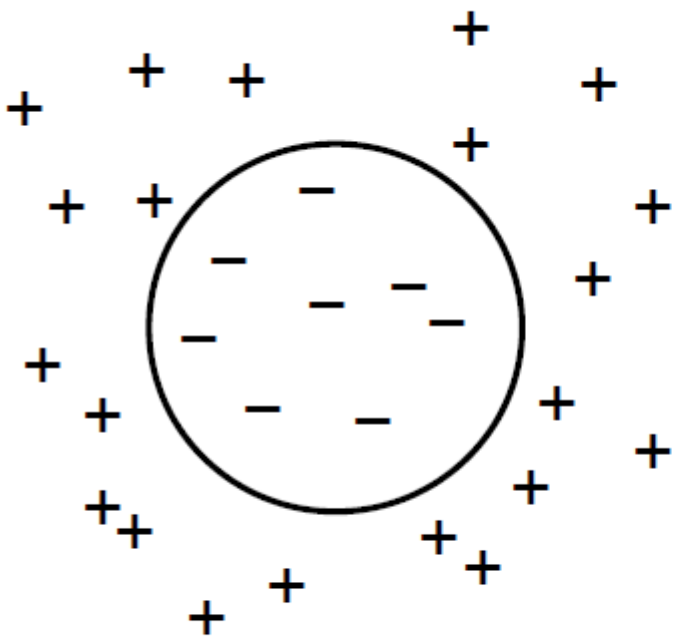
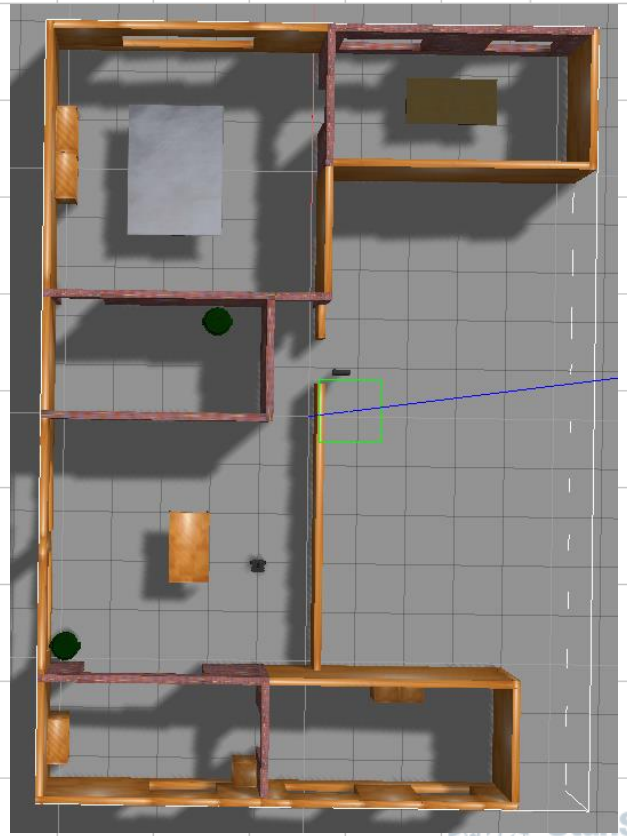
Generalized Definition



(a)

(b)

- Idea: why limit to linear functions
- Example:
  - Disc with radius 2 centered at origin:
- Solid representation:

Summary

- Define the world: $W \subseteq \mathbb{R}^2$ or $W \subseteq \mathbb{R}^3$
  - Two kinds of entites:
    1. Obstacles (permanently occupied) $O \subseteq W$
    2. Robots / moving bodies $R(t) \subseteq W$
- Objects defined in terms of finite set of **Unions**, **Intersections**, and **Set Differences**
  - Boundary: Edges and vertices, polygon specified by vertex sequence
  - Convex solid: Intersection of halfplanes
  - Non-convex solid: Represent as a union of convex polygons
- Determine if point lies within obstacle: use predicates
  - Single convex obstacle: "And" operator
  - Entire obstacle region: "Or" operator
- 3D
  - Polygons → Polyhedra
  - Boundary: Vertices & edges → Vertices, edges, & faces
  - Solid: Half-plane → Half-space
- No need to stick with pure linear functions